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Allocation of new growth between shoot, root and mycorrhiza in relation to carbon, nitrogen and phosphate supply: Teleonomy with maximum growth rate

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HIGHLIGHTS

- Expressions allocating new dry matter to shoot, root and mycorrhiza are derived which maximize growth rate.
- These demonstrate several key intuitive phenomena concerning resource sharing between plant components and associated mycorrhizae.
- The approach offers a methodology for introducing resource sharing between species into ecosystem models.
- Such teleonomic (optimal response) models may provide a valuable means of simulating allocation.
- This can avoid the circularity of empirical models and circumvent the complexities and uncertainties inherent in mechanistic approaches.

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ABSTRACT

Treating resource allocation within plants, and between plants and associated organisms, is essential for plant, crop and ecosystem modelling. However, it is still an unresolved issue. It is also important to consider quantitatively when it is efficient and to what extent a plant can invest profitably in a mycorrhizal association. A teleonomic model is used to address these issues. A six state-variable model giving exponential growth is constructed. This represents carbon (C), nitrogen (N) and phosphorus (P) substrates with structure in shoot, root and mycorrhiza. The shoot is responsible for uptake of substrate C, the root for substrates N and P, and the mycorrhiza also for substrates N and P. A teleonomic goal, maximizing proportional growth rate, is solved analytically for the allocation fractions. Expressions allocating new dry matter to shoot, root and mycorrhiza are derived which maximize growth rate. These demonstrate several key intuitive phenomena concerning resource sharing between plant components and associated mycorrhizae. For instance, if root uptake rate for phosphorus is equal to that achievable by mycorrhiza and without detriment to root uptake rate for nitrogen, then this gives a faster growing mycorrhizal-free plant. However, if root phosphorus uptake is below that achievable by mycorrhiza, then a mycorrhizal association may be a preferred strategy. The approach offers a methodology for introducing resource sharing between species into ecosystem models. Applying teleonomy may provide a valuable short-term means of modelling allocation, avoiding the circularity of empirical models, and circumventing the complexities and uncertainties inherent in mechanistic approaches. However it is subjective and brings certain irreducible difficulties with it.

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1. Introduction

There continue to be difficulties modelling dry matter allocation in plant growth and plant ecosystem models. The latter are becoming increasingly important with the growing concern with climate change and carbon sequestration. The problem is sufficiently challenging that many crop and plant ecosystem models use simple "outcome-level" empiricism, an approach that is unlikely to give useful insights and often leads to circularity. While mechanistic models represent the "holy grail", they may be unattainable at the present time. Arguably, an approach that could provide progress is to use "functionality" – assuming a goal which the system attempts to attain via means which exist because of adaptation. Apparent goal-seeking behaviour can only be obtained through mechanisms, which may or may not exist for that purpose. However, postulating goals

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and assuming that the means have evolved to fulfil those goals provides an alternative (but arguably short-term) approach to the allocation problem. It leads to models which are sometimes dubbed teleonomic or functional, or optimal response.

While here we develop teleonomic approaches, one should not forget the importance of mechanism and the many successes achieved by traditional scientific reductionism, although this approach can require great persistence and a long time horizon, both of which may not be easy to obtain. This situation is highlighted by a recent review of allocation (Poorter et al., 2011), which appears to be devoid of hypotheses or synthetic mathematical statements (Warren Wilson et al., 1986). To misquote Rutherford: "all real science is reductionist; the rest is just stamp collecting". Nevertheless, teleonomic relationships can contribute to the investigation of complex systems when too little may be known for reductionism to proceed confidently.

Some reviews of modelling allocation are by Franklin et al. (2012, who omit mechanistic methods), Dewar et al. (2009, they discuss teleonomic models of forest response), Thornley and France (2007,, pp. 8–9, 326–341), Lacointe (2000) (an excellent review), Thornley (1998a), Cannell and Dewar (1994), Mäkelä (1990, covers functional balance and pipe theory) and Wilson (1988). Early work in the area is often described by the term functional equilibria, e.g. White (1937), Brouwer (1962) and Davidson (1969). Functionality can equally well be described as goal-seeking or as teleonomy (Monod, 1974): we decide the function or goal, possibly guided by experimental data, or ideas about fitness or evolution. Indeed, much of allocation modelling straddles uneasily the fault line between mechanistic approaches and functional or teleonomic approaches, attempting to make progress in this difficult but important area. Hopefully the present teleonomic study can contribute to this debate. The particular teleonomic approach applied here was first suggested by Reynolds and Thornley (1982), and has been further discussed by Johnson and Thornley (1987) and Thornley and France (2007), pp. 328–333). The method adjusts allocation fractions so as to maximize growth rate. An important paper by Mäkelä and Sievänen (1987) gives a detailed comparison of the (mechanistic) transport-resistance and teleonomic approaches to C, N allocation in a shoot–root model, concluding significantly that the teleonomic model essentially is embedded in the mechanistic model. This is important because it demonstrates that apparently goal-seeking predictions can be generated by an objective mechanistic model. Dewar et al. (2009) review "optimal function" (or teleonomic) modelling in forests, and make recommendations for further work to progress this approach to modelling.

In this paper we revisit and extend a teleonomic model which maximizes proportional growth rate while allocating carbon (C) and nitrogen (N) to shoot and root. The extension includes phosphorus (P) as a third nutrient and mycorrhiza as a third possible compartment. Our objectives are first to develop teleonomic methodology for an extended system, second to see if it can provide new insights into plant-mycorrhizal associations, and last to sharpen up (make more quantitative) the questions in this area. The method may help provide submodels suitable for including in larger crop and ecosystem simulators.

2. Model and methods

2.1. Basic analysis

The model scheme is drawn in Fig. 1. Substrate C is taken up by the shoot, substrates N and P are taken up by both root and mycorrhiza. The five k shown in Fig. 1 are uptake parameters [Eq. (1)]. We study teleonomic allocation between shoot, root and mycorrhiza, maximizing overall growth rate of the association. This optimization, which determines allocation, is performed for particular values of the uptake parameters. The procedure is carried out for the model system in "steady-state" exponential growth, when all parts of the system, shoot, root and mycorrhiza, grow at the same proportional rate. Exponential growth is assumed for three reasons: first it permits a thorough algebraic analysis; second, arguably, early exponential growth is important for "fitness" or survival, partially justifying our particular teleonomic assumption; and last, empirical data indicate that early plant growth can be approximately exponential before processes such as self-shading, root competition and respiration set in (e.g. Causton and Venus, 1981, p. 39). The model is constructed so that exponential-growth solutions exist.

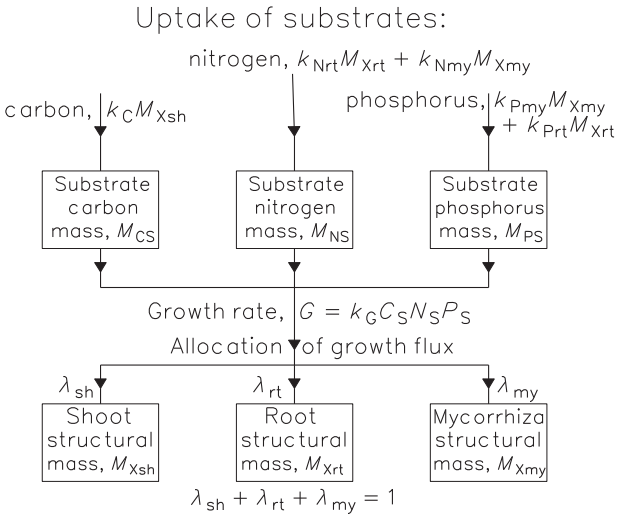


Fig. 1. Model scheme for carbon, nitrogen and phosphorus uptake, growth and allocation in a plant-mycorrhizal system. This comprises shoot (sh), root (rt) and mycorrhiza (my). There are six state variables as indicated by the boxes. Uptake fluxes of C, N, P substrates are associated with shoot, root, and mycorrhiza as shown [Eq. (1)]. Total structural dry matter growth rate, G [Eq. (6)], is allocated according to fractions, λ_{sh}, λ_{rt} and λ_{my} to shoot, root and mycorrhiza [Eq. (10)]. See Eq. (2) for substrate concentrations, C_S, N_S and P_S and Eq. (3) for total structural dry matter, M_X.

State variables, other variables and parameters with default values are listed in Table 1. There are six mass (M) state variables, shown in Fig. 1: three for the structural components (X) for shoot M_{Xsh}, root M_{Xrt} and mycorrhiza M_{Xmy}, with units of kg structural dry matter; and three for the substrate pools (S), M_{CS}, M_{NS} and M_{PS}, with units of kg carbon (C), nitrogen (N) and phosphorus (P) substrates. The substrates are assumed to be equally accessible to all parts of the system.

The uptake rates of C, N and P substrates, U_C, U_N and U_P (units: kg substrate C, N, P d^{-1}) are

$$U_C = k_C M_{Xsh}, \quad U_N = k_{Nrt}M_{Xrt} + k_{Nmy}M_{Xmy}, \quad U_P = k_{Pmy}M_{Xmy} + k_{Prt}M_{Xrt},$$

$$k_C = 0.2 \text{ kg substrate C (kg shoot structural dry matter)}^{-1}\ \text{d}^{-1},$$

$$k_{Nrt} = 0.05 \text{ kg substrate N (kg root structural dry matter)}^{-1}\ \text{d}^{-1},$$

$$k_{Nmy} = 0.025 \text{ kg substrate N (kg mycorrhiza structural dry matter)}^{-1}\ \text{d}^{-1},$$

$$k_{Pmy} = 0.02\ \text{kg substrate P (kg mycorrhizal structural dry matter)}^{-1}\ \text{d}^{-1},$$

$$k_{Prt} = 0.01\ \text{kg substrate P (kg root structural dry matter)}^{-1}\ \text{d}^{-1}, \qquad (1)$$

with uptake parameters k_C, k_{Nrt}, k_{Nmy}, k_{Pmy} and k_{Prt}. Note that here $k_{Nrt} > k_{Nmy}$ and $k_{Pmy} > k_{Prt}$: the root is better than the mycorrhiza at N uptake, and vice versa for P uptake so that the optimum system for growth comprises root and mycorrhiza. These are illustrative values, not based directly on empirical evidence, except that they support acceptable rates of plant growth. In the simulations a range is k values is considered.

Table 1
State variables (with initial values), parameters (with default values) and definitions: equation numbers where the items are introduced are indicated.

State variable	Description	Initial ($t=0$) value
(a)		
M_{CS}, M_{NS}, M_{PS}	Mass of substrate C, N, P [Eq. (12)]	0.025, 0.01, 0.005 kg substrate C, N, P
M_{Xsh}, M_{Xrt}, M_{Xmy}	Mass of structural dry matter in shoot, root, mycorrhiza [Eq. (13)]	0.2, 0.3, 0.5 kg structural dry matter
(b)		
Parameter + default value	Description	Units
$f_{CX}=0.4$, $f_{NX}=0.02$, $f_{PX}=0.005$	Fractions of C, N, P in structural dry matter [Eq. (8)]	kg structural C, N, P (kg structural dry matter)$^{-1}$
$J_{CS}=J_{NS}=J_{PS}=7.7777\times10^{6}$	Uptake inhibition constants [Eq. (43)]	kg substrate C, N, P (kg structural dry matter)$^{-1}$
$k_C=0.2$, $k_{Nrt}=0.05$, $k_{Pmy}=0.02$, $k_{Nmy}=0.025$, $k_{Prt}=0.01$	Uptake rate constants for substrate C, N, P [Eq. (1)]	kg substrate C, N, P (kg structural dry matter)$^{-1}$ d^{-1}
$k_G=200{,}000$	Structural growth rate constant [Eq. (6)]	($[C_S]$ $[N_S]$ $[P_S]$)$^{-1}$ d^{-1}
$q=0.5$	Control parameter for teleonomic allocation [Eq. (31)]	
(c)		
Other variables	Description	Units
C_S, N_S, P_S	Concentrations of C, N, P substrates [(2)]	kg substrate C, N, P (kg structural dry matter)$^{-1}$
f_{Xsh}, f_{Xrt}, f_{Xmy}	Fractions of structural dry matter in shoot, root, mycorrhiza [Eq. (4)]	
G	Growth rate of structural dry matter [Eq. (6)]	kg structural dry matter d^{-1}
M_X	Total structural dry matter [Eq. (3)]	kg structural dry matter
U_C, U_N, U_P	Uptake rates of C, N, P substrates [Eq. (1)]	kg substrate C, N, P d^{-1}
$\lambda_{sh}=0.4$, $\lambda_{rt}=0.35$, $\lambda_{my}=0.25$	Allocation fractions (constant, or variable with teleonomy) [Eqs. (11), (13), and (31)]	
μ_{CS}, μ_{NS}, μ_{PS}, μ_{Xsh}, μ_{Xsh}, μ_{Xsh}, μ_{Xs} μ	Proportional rates of growth of state variables and total structural dry matter [Eqs. (14) and (15)]; and their common value	d^{-1}

Substrate "concentrations" of carbon, nitrogen and phosphorus, C_S, N_S and P_S [units: kg substrate C, N, P (kg structural dry matter)$^{-1}$] are defined by

$$C_S=\frac{M_{CS}}{M_X},\quad N_S=\frac{M_{NS}}{M_X},\quad P_S=\frac{M_{PS}}{M_X} \tag{2}$$

These are not true concentrations, although we shall refer to them as concentrations. Total structural matter, M_X (kg) and its rate of change are [using Eq. (13) below]

$$M_X=M_{Xsh}+M_{Xrt}+M_{Xmy},$$
$$\frac{dM_X}{dt}=\frac{dM_{Xsh}}{dt}+\frac{dM_{Xrt}}{dt}+\frac{dM_{Xmy}}{dt} \tag{3}$$

Define structural fractions [from Eq. (3)] with

$$f_{Xsh}=\frac{M_{Xsh}}{M_X},\ f_{Xrt}=\frac{M_{Xrt}}{M_X},\ f_{Xmy}=\frac{M_{Xmy}}{M_X};\quad f_{Xsh}+f_{Xrt}+f_{Xmy}=1. \tag{4}$$

At time zero, $t=0$ d, then [from Eqs. (4) and (13)]

$$f_{Xsh}(t=0)=0.2,\quad f_{Xrt}(t=0)=0.3,\quad f_{Xmy}(t=0)=0.5. \tag{5}$$

These initial values are set far from the optimum so that the dynamics of moving to "steady-state" exponential growth with constant allocation or alternatively with optimum allocation [Eq. (11)] are more easily observed (Figs. 3 and 4).

Structural growth rate, G (kg structural dry matter d^{-1}), is assumed to be

$$G=k_G M_X C_S N_S P_S,$$
$$k_G=200{,}000\ ([C_S][N_S][P_S])^{-1}\ \text{d}^{-1} \tag{6}$$

G is proportional to total structural dry matter [M_X, Eq. (3)] times the product of the substrate concentrations [Eq. (2)], with constant k_G [C_S] etc. denotes the units of the variable, C_S [see before Eq. (2)]. The theoretical justification of Eq. (6) is that growth must depend on the availability of substrates and be zero if substrates are not available. An empirical justification is shown in Fig. 3 of Thornley and Hurd (1974) for young tomato plants, based on data presented in Hurd and Thornley (1974).

In the absence of senescence or other loss of structural dry matter (Fig. 1), structural growth rate equals the rate of increase in structural matter:

$$\frac{dM_X}{dt}=G\quad\text{and}\quad \mu=\frac{1}{M_X}\frac{dM_X}{dt}=\frac{G}{M_X}=k_G C_S N_S P_S \tag{7}$$

μ (d^{-1}) is the proportional growth rate of total structural dry matter. Again, using illustrative values, if the substrate concentrations [Eq. (2)] have values of $C_S=0.025$, $N_S=0.01$ and $P_S=0.005$ kg substrate C, N, P (kg structural dry matter)$^{-1}$ [as given by the initial values in Eq. (12) when $M_X(t=0)=1$ kg, Eq. (13)], then $\mu=200{,}000\times0.25\times0.01\times0.005=0.25$ d^{-1}. Assuming that structural dry matter is 40% carbon [Eq. (8)], then our substrate concentrations of $C_S=0.025$, $N_S=0.01$ and $P_S=0.005$ are equivalent to a substrate C: structural C ratio of 0.0625 (0.025/0.4), substrate N: structural C ratio of 0.025 (0.01/0.4), and a substrate P: structural C ratio of 0.0125 (0.005/0.4).

Assume that the fractional C, N and P contents of structural dry matter are constant and the same for shoot, root and mycorrhiza:

$$f_{CX}=0.400,\ f_{NX}=0.025,\ f_{PX}=\ 0.005. \tag{8}$$

Units are kg structural C (N, P) per kg of structural dry matter. This gives elemental mass ratios in structural dry matter of C: 1 kg C/kg C (0.4/0.4); N: 0.0625 kg N/kg C (0.025/0.4); P: 0.0125 (0.005/0.4) kg P/kg C. The assumption of equality is clearly an approximation; it is not essential but it simplifies the algebra; Landis and Fraser (2008) assume C, N, P biomass fractions for the plant of 0.33, 0.1, 0.01, and for the mycorrhiza of 0.5, 0.1 0.01; however, measured mineral contents are highly variable (e.g. Jeffrey, 1988, p. 185). Growth at rate G [Eq. (6)] requires C, N and P substrates at rates (kg C, N, P substrate d^{-1})

$$f_{CX}G, f_{NX}G, f_{PX}G. \tag{9}$$

New growth at rate G is allocated to shoot (sh), root (rt) and mycorrhiza (my) with allocation fractions λ_{sh}, λ_{rt}, λ_{my} (Fig. 1):

$$G\lambda_{sh},\ G\lambda_{rt},\ G\lambda_{my} \text{ with } \lambda_{sh}+\lambda_{rt}+\lambda_{my}=1. \tag{10}$$

Allocation fractions, λ, are either, determined according to a teleonomic algorithm [Eqs. (31) and (41)] which maximizes proportional growth rate, μ [Eq. (7)], or, are given constant values. Thus

$$\begin{aligned}&\text{either } [\lambda_{sh}=\lambda_{sh}(\text{teleonomic}),\ \lambda_{rt}=\lambda_{rt}(\text{teleonomic}),\ \lambda_{my}=\lambda_{my}(\text{teleonomic})],\\ &\text{or } [\lambda_{sh}=\lambda_{sh}(\text{constant}),\ \lambda_{rt}=\lambda_{rt}(\text{constant}),\ \lambda_{my}=\lambda_{my}(\text{constant}),\ \text{where}\\ &\lambda_{sh}(\text{constant})=0.4,\ \lambda_{rt}(\text{constant})=0.35,\ \lambda_{my}(\text{constant})=0.25].\end{aligned} \tag{11}$$

The differential equations for the three substrate state variables (Fig. 1) are [using Eqs. (1), (10) and (6)]

$$\begin{aligned}&\frac{dM_{CS}}{dt}=k_C M_{Xsh}-f_{CX}k_G M_X C_S N_S P_S,\quad M_{CS}(t=0)=0.025 \text{ kg C substrate},\\ &\frac{dM_{NS}}{dt}=k_{Nrt}M_{Xrt}+k_{Nmy}M_{Xmy}-f_{NX}k_G M_X C_S N_S P_S,\quad M_{NS}(t=0)=0.01 \text{ kg N substrate},\\ &\frac{dM_{PS}}{dt}=k_{Pmy}M_{Xmy}+k_{Prt}M_{Xrt}-f_{PX}k_G M_X C_S N_S P_S,\quad M_{PS}(t=0)=0.005 \text{ kg P substrate}.\end{aligned} \tag{12}$$

With an initial ($t=0$ d) value of $M_X=1$ kg structural dry matter [Eqs. (13) and (3)], the initial substrate concentrations [Eq. (2)] are 0.025, 0.01 and 0.005 kg substrate C, N, P (kg structural dry matter)$^{-1}$ respectively.

The differential equations for the three structural dry matter state variables (Fig. 1) are

$$\begin{aligned}&\frac{dM_{Xsh}}{dt}=\lambda_{sh}k_G M_X C_S N_S P_S,\quad M_{Xsh}(t=0)=0.2 \text{ kg structural dry matter},\\ &\frac{dM_{Xrt}}{dt}=\lambda_{rt}k_G M_X C_S N_S P_S,\quad M_{Xrt}(t=0)=0.3 \text{ kg structural dry matter},\\ &\frac{dM_{Xmy}}{dt}=\lambda_{my}k_G M_X C_S N_S P_S,\quad M_{Xmy}(t=0)=0.5 \text{ kg structural dry matter},\end{aligned} \tag{13}$$

λ_{sh}, λ_{rt} and λ_{my} are the dynamic allocation fractions [Fig. 1, Eq. (10)], which may be ascribed constant values [Eq. (11)], or may be varied [Eqs. (31) and (41)] so as to give maximum growth rate, μ [Eq. (7)], for the system. In Eq. (13), initial values of structural dry matter fractions [Eqs. (5) and (4)] are deliberately made rather different from the constant λ values and the optimized values, which determine the equilibrium values of Eq. (4): this enables the transient to be easily visible.

2.2. Exponential growth

In the six differential equations [Eqs. (12) and (13)], all terms scale linearly with increasing plant size, so that exponential growth solutions are a possible outcome, which are obtained in this section. Define proportional growth rates (d^{-1}) of the state variables:

$$\begin{aligned}&\mu_{CS}=\tfrac{1}{M_{CS}}\tfrac{dM_{CS}}{dt},\quad \mu_{NS}=\tfrac{1}{M_{NS}}\tfrac{dM_{NS}}{dt},\quad \mu_{PS}=\tfrac{1}{M_{PS}}\tfrac{dM_{PS}}{dt},\\ &\mu_{Xsh}=\tfrac{1}{M_{Xsh}}\tfrac{dM_{Xsh}}{dt},\quad \mu_{Xrt}=\tfrac{1}{M_{NS}}\tfrac{dM_{Xrt}}{dt},\quad \mu_{Xmy}=\tfrac{1}{M_{Xmy}}\tfrac{M_{Xmy}}{dt}.\end{aligned} \tag{14}$$

The proportional growth rates are intensive variables (like temperature and concentrations, they do not scale with plant size; extensive variables are quantities like amount, mass, area and volume, which depend on the size of the system). The proportional growth rates are constant during exponential growth, and then are all equal to μ. With Eq. (7):

$$\mu=k_G C_S N_S P_S. \tag{15}$$

The structural fractions, f_{Xsh}, f_{Xrt}, f_{Xmy} [Eq. (3)], are constant during exponential growth. For a small time interval, Δt, we must have

$$\frac{\Delta M_{Xsh}}{M_{Xsh}}=\frac{\Delta M_{Xrt}}{M_{Xrt}}=\frac{\Delta M_{Xmy}}{M_{Xmy}}=\frac{\Delta M_X}{M_X}=\mu\,\Delta t. \tag{16}$$

Therefore (because $\Delta M_{Xsh}/\Delta M_X=M_{Xsh}/M_X$ etc.), in exponential growth

$$\lambda_{sh}=f_{Xsh},\quad \lambda_{rt}=f_{Xrt},\quad \lambda_{my}=f_{Xmy}. \tag{17}$$

In exponential growth, dynamic allocation fractions λ [Eq. (10)] equal the partitioning fractions f_X [Eq. (4)] for shoot, root and mycorrhiza.

For exponential growth, the six differential equations [Eqs. (12) and (13)] become [with Eqs. (17) and (15) and a minor re-arrangement]

$$\begin{aligned}&\mu=k_G C_S N_S P_S,\\ &\mu(C_S+f_{CX})=k_C\lambda_{sh},\\ &\mu(N_S+f_{NX})=k_{Nrt}\lambda_{rt}+k_{Nmy}\lambda_{my},\\ &\mu(P_S+f_{PX})=k_{Pmy}\lambda_{my}+k_{Prt}\lambda_{rt},\\ &\lambda_{sh}+\lambda_{rt}+\lambda_{my}=1.\end{aligned} \tag{18}$$

Parameters whose values are assumed are f_{CX}, f_{NX}, f_{PX}, k_G, k_C, k_{Nrt}, k_{Nmy}, k_{Pmy} and k_{Prt} [Eqs. (8), (6), (1)]. Values may be assigned to the allocation/partitioning fractions, λ_{sh}, λ_{rt}, λ_{my} [Eqs. (4), (17), (13) with $f_X=\lambda$] so that they sum to unity. This leaves four equations for four unknowns: μ, C_S, N_S and P_S.

The next step is to derive from Eq. (18) a polynomial $f(\mu)$ in μ, the proportional rate of growth [Eqs. (14) and (15)]:

$$\begin{aligned}&C_S=\frac{k_C\lambda_{sh}}{\mu}-f_{CX},\quad N_S=\frac{k_{Nrt}\lambda_{rt}+k_{Nmy}\lambda_{my}}{\mu}-f_{NX},\\ &P_S=\frac{k_{Pmy}\lambda_{my}+k_{Prt}\lambda_{my}}{\mu}-f_{PX},\\ &\mu=k_G\left(\frac{k_C\lambda_{sh}}{\mu}-f_{CX}\right)\left(\frac{k_{Nrt}\lambda_{rt}+k_{Nmy}\lambda_{my}}{\mu}-f_{NX}\right)\\ &\qquad\times\left(\frac{k_{Pmy}\lambda_{my}+k_{Prt}\lambda_{my}}{\mu}-f_{PX}\right),\\ &\mu^4=k_G(k_C\lambda_{sh}-\mu f_{CX})(k_{Nrt}\lambda_{rt}+k_{Nmy}\lambda_{my}-\mu f_{NX})\\ &\qquad\times(k_{Pmy}\lambda_{my}+k_{Prt}\lambda_{my}-\mu f_{PX}),\\ &0=\mu^4+\mu^3 k_G f_{CX}f_{NX}f_{PX}\\ &\quad-\mu^2 k_G[k_C\lambda_{sh}f_{NX}f_{PX}+(k_{Nrt}\lambda_{rt}+k_{Nmy}\lambda_{my})f_{CX}f_{PX}\\ &\quad+(k_{P,\,my}\lambda_{my}+k_{P,\,rt}\lambda_{my})f_{CX}f_{NX}]\\ &\quad+\mu k_G[(f_{CX}(k_{Nrt}\lambda_{rt}+k_{Nmy}\lambda_{my})(k_{Pmy}\lambda_{my}+k_{Prt}\lambda_{my})\\ &\quad+f_{NX}k_C\lambda_{sh}(k_{Pmy}\lambda_{my}+k_{Prt}\lambda_{my})\\ &\quad+f_{PX}k_C\lambda_{sh}(k_{Nrt}\lambda_{rt}+k_{Nmy}\lambda_{my})]\\ &\quad-k_G k_C\lambda_{sh}(k_{Nrt}\lambda_{rt}+k_{Nmy}\lambda_{my})(k_{Pmy}\lambda_{my}+k_{Prt}\lambda_{my}),\\ &0=a\mu^4+b\mu^3+c\mu^2+d\mu+e.\\ &f(\mu)=a\mu^4+b\mu^3+c\mu^2+d\mu+e.\end{aligned} \tag{19}$$

The coefficients, a, b, …, e, are defined in the preceding equation (i.e. $a=1$, $b=k_G f_{CX}f_{NX}f_{PX}$ etc.). $f(\mu)$ defines the quartic polynomial in μ. $f(\mu)$ can be regarded as a function of the allocation fractions λ.

The quartic equation [$0=f(\mu)$] can be solved for μ, and thence for substrate concentrations, C_S, N_S and P_S using the first three of Eq. (19). Some general features of the solutions can be noted by inspection.

There are always at least two real solutions, one with $\mu<0$ and one with $\mu\geq 0$ because $f(\mu=0)<0$ (e is negative) and $f(\mu\to\pm\infty)\to+\infty$. Looking at the 4th statement ($\mu=\ldots$) in Eq. (19),

it can be seen that for large k_G there are three possible μ-asymptotes:

$$\begin{aligned}\mu_C &= \frac{k_C(0.2)\lambda_{sh}(0.4)}{f_{CX}(0.4)} = 0.2,\\ \mu_N &= \frac{k_{Nrt}(0.05)\lambda_{rt}(0.35)+k_{Nmy}(0.025)\lambda_{my}(0.25)}{f_{NX}(0.02)} = 1.1875,\\ \mu_P &= \frac{k_{Pmy}(0.02)\lambda_{my}(0.25)+k_{Prt}(0.01)\lambda_{rt}(0.35)}{f_{PX}(0.005)} = 1.7,\\ \mu(\text{actual}) &= 0.1999.\end{aligned} \tag{20}$$

Values inserted here are for default parameters (Table 1) with constant allocation [Eq. (11)]. The actual growth rate (bottom line) is slightly less than the smallest of the limiting growth rates which in this case is C assimilation. If the λ are adjusted to maximize the growth rate [Eqs. (30) and (31), see Fig. 4 and text thereafter], then these quantities change to

$$\begin{aligned}&\mu_C = 0.3715,\ \ \mu_N = 0.5466,\ \ \mu_P = 0.6679 \text{ and } \mu(\text{actual}) = 0.3445,\\ &\text{with } \lambda_{sh} = 0.743,\ \ \lambda_{rt} = 0.180,\ \ \lambda_{my} = 0.0769.\end{aligned} \tag{21}$$

The shoot fraction in Eq. (21) is much increased relative to Eqs. (20) and (11) where $\lambda_{sh}=0.4$ so that C assimilation is far less limiting than with the default parameters [Eq. (20)]. Remember that Eq. (17) are satisfied in exponential growth.

For the default parameters (Table 1), there are only two real roots for growth rate μ with one of these >0 and biologically meaningful. If the phosphate uptake rate constants, k_{Pmy} and k_{Prt} [Eq. (1)], are changed from $k_{Pmy}=0.02$ to 0.0002 and $k_{Prt}=0.01$ to0 kg *P* (kg mycorhhizal structural dry matter)$^{-1}$ d^{-1} then there are four real roots with only the lowest positive root being biologically meaningful. If C, N or P uptake is zero, that is, the parameters k_C, or k_{Nrt} and k_{Nmy}, or k_{Pmy} and k_{Prt}, is zero [Eq. (1)], then e is zero and $\mu=0$ is a solution, giving zero growth.

The biologically significant solution to the quartic in Eq. (19) is most easily obtained by simply integrating the differential Eqs. (12) and (13) until the system is in a state of "steady-state" exponential growth. This provides the proportional growth rate μ for any given set of parameters. Before we proceed to consider maximizing growth rate μ in relation to the allocation fractions λ_{sh}, λ_{rt}, λ_{my} [Eq. (10)], it is helpful to consider how, numerically, proportional growth rate μ depends on the allocation fractions – that is, constant values are assigned to the allocation fractions and the dynamic equations are integrated until μ is constant.

In Fig. 2A, maximum growth rate is achieved when both root and mycorrhiza are present, as the root is best at N uptake whereas the mycorrhiza is best at P uptake [Eq. (1)]. Note that, for a shoot allocation fraction of $\lambda_{sh}=0.7$ (this is quite close to the optimum value of $\lambda_{sh}=0.74$, below and Fig. 4C) growth rate is relatively insensitive to the allocation of growth between root and mycorrhiza. It follows that "optimum" allocation may not be a well-defined quantity (this raises doubts about the ecological concern for fitness or an optimum response – general physico-chemical constraints and accident may be more important in determining evolutionary outcomes). Moreover, as shoot allocation fraction λ_{sh} becomes increasing limiting at 0.6 and 0.4, the response of proportional growth rate μ to changing mycorrhizal/root allocation becomes negligible. It seems likely that, under certain conditions allocation is unimportant and may therefore be expected to be highly variable. In Fig. 2B, the root is better than the mycorrhiza at both N and P uptake, so that the optimum allocation is with no mycorrhiza. In Fig. 2C, root and mycorrhiza are equally good at taking up N and P so that the allocation between root and mycorrhiza does not matter. Finally in Fig. 2D, the mycorrhiza is best at both N and P uptake so that maximum growth is attained (within the context of this model) with zero root allocation.

2.3. Optimum allocation to give maximum exponential growth rate

This is now considered analytically, giving an algorithm which can be used in plant growth simulators to determine allocation.

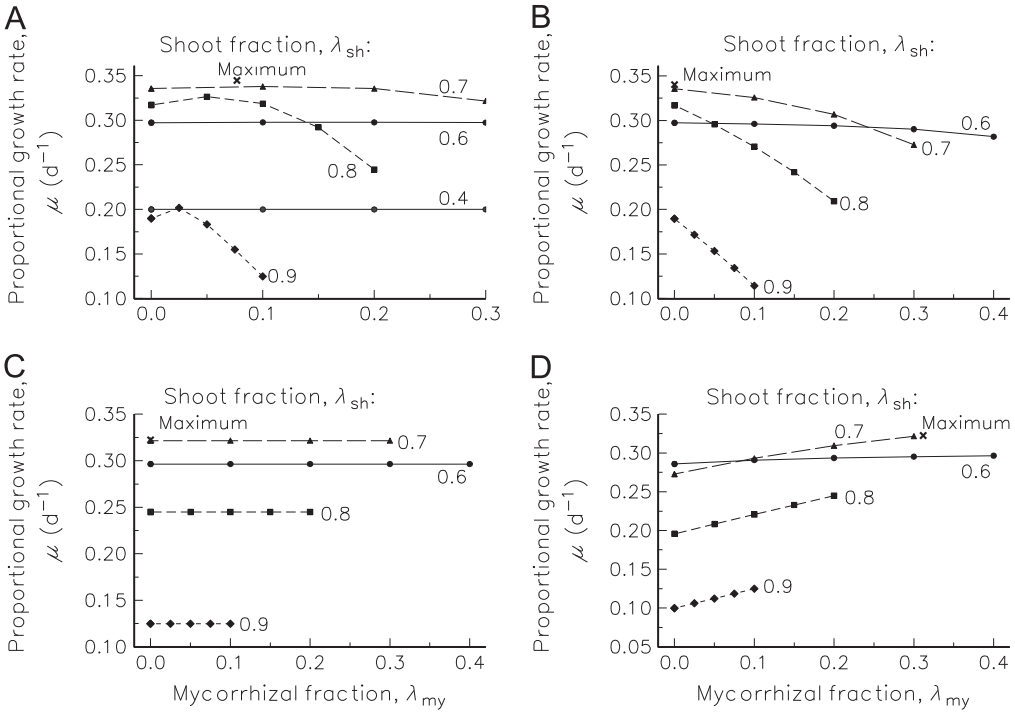


Fig. 2. Dependence of proportional growth rate μ [Eq. (15)] on allocation fractions λ_{sh}, λ_{rt}, λ_{my} [Fig. 1, Eq. (10)]. For a given value of λ_{sh}, mycorrhizal fraction λ_{my} is varied from 0 to $1-\lambda_{sh}$; $\lambda_{rt}=1-\lambda_{sh}-\lambda_{my}$, although not all these values are plotted. Eqs. (12) and (13) are integrated keeping the allocation fractions constant [Eq. (11)] until a constant exponential growth rate is achieved. This is done for several values of the uptake parameters [Eq. (1)]. (×) marks the point of maximum growth rate. (A) Default parameters (Table 1) – root is better than the mycorrhiza at N uptake, mycorrhiza is better at P uptake than the root ((×) is at $\mu=0.345$, $\lambda_{sh}=0.743$, $\lambda_{rt}=0.180$, $\lambda_{my}=0.077$). (B) $k_{Pmy}=0.006$ so that the root is better than the mycorrhiza at both N and P uptake ((×) is at $\mu=0.340$, $\lambda_{sh}=0.735$, $\lambda_{rt}=0.265$, $\lambda_{my}=0.0$). (C) $k_{Pmy}=0.02$, $k_{Prt}=0.02$, $k_{Nrt}=0.025$, $k_{Nmy}=0.025$ so that root and mycorrhiza are equally effective at both N and P uptake ((×) is at $\mu=0.322$, $\lambda_{sh}=0.688$, $\lambda_{rt}+\lambda_{my}=0.312$). (D) $k_{Nrt}=0.02$ so that the mycorrhiza is better than the root at both N and P uptake ((×) is at $\mu=0.322$, $\lambda_{sh}=0.688$, $\lambda_{rt}=0.0$, $\lambda_{my}=0.312$).

As shown in Fig. 2, the exponential growth rate μ obtained from Eq. (19) depends on the values assigned to the allocation fractions for shoot, root and mycorrhiza, λ_{sh}, λ_{rt} and λ_{my} [Eqs. (10) and (17)]. Now assume that the plant is able to adjust these fractions so that μ is maximized. This goal is ascribed to the system with no concern for mechanism. What are the consequences of this assumption?

Eq. (18) can be written (eliminating λ_{my} using the fifth equation) (we proceed by eliminating λ_{my} and λ_{rt}, although the same results are obtained by eliminating different pairs of the three allocation fractions)

$$\begin{aligned}&\mu\left(\tfrac{C_S+f_{CX}}{k_C}\right)=\lambda_{sh},\\&\mu(N_S+f_{NX})=k_{Nrt}\lambda_{rt}+k_{Nmy}(1-\lambda_{sh}-\lambda_{rt}),\\&\mu(P_S+f_{PX})=k_{Pmy}(1-\lambda_{sh}-\lambda_{rt})+k_{Prt}\lambda_{rt},\\&\mu=k_G C_S N_S P_S.\end{aligned}\tag{22}$$

From the 3rd of Eq. (22), λ_{rt} is

$$\lambda_{rt}=\frac{\mu(P_S+f_{PX})-k_{Pmy}(1-\lambda_{sh})}{k_{Prt}-k_{Pmy}}.\tag{23}$$

Now eliminate λ_{rt} from the 2nd of Eq. (22):

$$\begin{aligned}&\mu(C_S+f_{CX})=\lambda_{sh}k_C,\\&\mu(N_S+f_{NX})-k_{Nmy}(1-\lambda_{sh})=(k_{Nrt}-k_{Nmy})\left(\tfrac{\mu(P_S+f_{PX})-k_{Pmy}(1-\lambda_{sh})}{k_{Prt}-k_{Pmy}}\right),\\&\mu=k_G C_S N_S P_S.\end{aligned}\tag{24}$$

Differentiate Eq. (24) with respect to λ_{sh}, denoting the differential of x with respect to $d/d\lambda_{sh}$ by x' (that is, $dx/d\lambda_{sh}=x'$):

$$\begin{aligned}&\mu' C_S+\mu' f_{CX}+\mu C_s'=k_C,\\&\mu'(N_S+f_{NX})+\mu C_s'+k_{Nmy}=\left(\tfrac{k_{Nrt}-k_{Nmy}}{k_{Prt}-k_{Pmy}}\right)[\mu'(P_S+f_{PX})+\mu P_s'+k_{Pmy}],\\&\mu'=k_G(C_s' N_S P_S+C_S C_s' P_S+C_S N_S P_s').\end{aligned}\tag{25}$$

Maximum growth rate $\mu=\tilde{\mu}$ is obtained when $\mu'=0$. Use $\tilde{x}$ to denote the value of variable x at the maximum. From Eq. (25):

$$\begin{aligned}&\tfrac{\tilde{\mu}\tilde{C}_s'}{k_C}=1,\\&\tilde{\mu}\tilde{N}_s'+k_{Nmy}=\tfrac{k_{Nrt}-k_{Nmy}}{k_{Prt}-k_{Pmy}}(\tilde{\mu}\tilde{P}_s'+k_{Pmy}),\\&0=\tilde{C}_s'\tilde{N}_S\tilde{P}_S+\tilde{C}_S\tilde{N}_s'\tilde{P}_S+\tilde{C}_S\tilde{N}_S\tilde{P}_s'.\end{aligned}\tag{26}$$

Eliminating $\tilde{\mu}$ from the first two of Eq. (26) and dividing the last equation by $\tilde{C}_S\tilde{N}_S\tilde{P}_S$:

$$\begin{aligned}&\tfrac{\tilde{C}_s'}{k_C}\left(k_{Nmy}-k_{Pmy}\tfrac{k_{Nrt}-k_{Nmy}}{k_{Prt}-k_{Pmy}}\right)+\tilde{N}_s'-\tilde{P}_s'\left(\tfrac{k_{Nrt}-k_{Nmy}}{k_{Prt}-k_{Pmy}}\right)=0,\\&\Rightarrow\tfrac{\tilde{C}_s'}{k_C}\left(\tfrac{k_{Nmy}k_{Prt}-k_{Nrt}k_{Pmy}}{k_{Prt}-k_{Pmy}}\right)+\tilde{N}_s'-\tilde{P}_s'\left(\tfrac{k_{Nrt}-k_{Nmy}}{k_{Prt}-k_{Pmy}}\right)=0,\\&\Rightarrow\tfrac{\tilde{C}_s'}{k_C}+\tilde{N}_s'\tfrac{k_{Pmy}-k_{Prt}}{k_{Nrt}k_{Pmy}-k_{Nmy}k_{Prt}}+\tilde{P}_s'\tfrac{k_{Nrt}-k_{Nmy}}{k_{Nrt}k_{Pmy}-k_{Nmy}k_{Prt}}=0,\\&0=\tfrac{\tilde{C}_s'}{\tilde{C}_S}+\tfrac{\tilde{N}_s'}{\tilde{N}_S}+\tfrac{\tilde{P}_s'}{\tilde{P}_S}.\end{aligned}\tag{27}$$

The last two equations of Eq. (27) are satisfied simultaneously only if

$$\frac{\tilde{C}_S}{k_C}=\frac{\tilde{N}_S}{(k_{Nrt}k_{Pmy}-k_{Nmy}k_{Prt})/(k_{Pmy}-k_{Prt})}=\frac{\tilde{P}_S}{(k_{Nrt}k_{Pmy}-k_{Nmy}k_{Prt})/(k_{Nrt}-k_{Nmy})}.\tag{28}$$

Define

$$k_N'=\frac{k_{Nrt}k_{Pmy}-k_{Nmy}k_{Prt}}{k_{Pmy}-k_{Prt}},\quad k_P'=\frac{k_{Nrt}k_{Pmy}-k_{Nmy}k_{Prt}}{k_{Nrt}-k_{Nmy}}\tag{29}$$

Therefore

$$\frac{\tilde{C}_S}{k_C}=\frac{\tilde{N}_S}{k_N'}=\frac{\tilde{P}_S}{k_P'}.\tag{30}$$

The two equations in Eq. (30) must be satisfied by the adjusted allocation fractions in order to give maximum growth rate.

The final step in the optimization scheme is to invent a dynamic allocation algorithm which forces the system into a situation where Eq. (30) and at the same time Eq. (17) for exponential growth are satisfied. One way in which the maximum growth-rate situation can be attained is by assuming that the current dynamic allocation fractions, λ_{sh}, λ_{rt} and λ_{my} [Eq. (11)] are given by the current variables of the system according to [use Eqs. (4), (3) and (2) to get back to the state variables]

$$\begin{aligned}&\lambda_{sh}(\text{teleonomic})=\frac{f_{Xsh}(k_C/C_S)^q}{f_{Xsh}(k_C/C_S)^q+f_{Xrt}(k_N'/N_S)^q+f_{Xmy}(k_P'/P_S)^q},\\&\lambda_{rt}(\text{teleonomic})=\frac{f_{Xrt}(k_N'/N_S)^q}{f_{Xsh}(k_C/C_S)^q+f_{Xrt}(k_N'/N_S)^q+f_{Xmy}(k_P'/P_S)^q},\\&\lambda_{my}(\text{teleonomic})=\frac{f_{Xmy}(k_P'/P_S)^q}{f_{Xsh}(k_C/C_S)^q+f_{Xrt}(k_N'/N_S)^q+f_{Xmy}(k_P'/P_S)^q},\\&q=0.5.\end{aligned}\tag{31}$$

When Eq. (30) are satisfied, then Eq. (31) give results that satisfy Eq. (17). q is an empirical dimensionless parameter that determines the dynamics of approaching maximum growth rate (e.g. $q=0$, 0.5, 1, …). If $q=0$, then $\lambda_{sh}=f_{Xsh}$, etc. and there is no teleonomic goal-seeking: the structural fractions [Eq. (4)] do not change from their initial values. Studies of allosteric proteins reveal the subtlety of what is possible in terms of regulation (Tsai et al., 2009; Monod et al., 1965; Savageau, 1995), and such or similar equations may well be within reach of feasible biological mechanisms. We assume here that q can have non-integral values, which may be somewhat inconsistent with the theory of allosteric proteins.

However, we note that using these equations does not necessarily satisfy the biological constraints that the allocation fractions [Eq. (10)] must lie between zero and unity and only positive values of the substrate concentrations in Eqs. (30) and (2) are allowed. The difficulty arises when k_N' and k_P' [Eq. (29)] take negative values. Defining

$$k_{Prt}=\xi k_{Pmy},\quad k_{Nmy}=\zeta k_{Nrt}.\tag{32}$$

Eq. (29) become

$$\begin{aligned}&k_N'=\frac{k_{Nrt}k_{Pmy}-\zeta k_{Nrt}\xi k_{Pmy}}{k_{Pmy}-\zeta k_{Pmy}}=k_{Nrt}\tfrac{1-\xi\zeta}{1-\zeta},\\&k_P'=\frac{(k_{Nmy}/\zeta)k_{Pmy}-k_{Nmy}\xi k_{Pmy}}{(k_{Nmy}/\zeta)-k_{Nmy}}=k_{Pmy}\tfrac{1-\xi\zeta}{1-\zeta}.\end{aligned}\tag{33}$$

The "normal" situation is considered to be where k_N' and k_P' are positive, $\xi<1$ (the root is less efficient at P uptake than the mycorrhiza) and $\varsigma<1$ (the mycorrhiza is less efficient at N uptake than the root). However k_N' and k_P' are both negative if

$$\begin{aligned}&[\xi\zeta>1\text{ and }\zeta<1]\,(a)\\&\text{or}\\&[\xi\zeta<1\text{ and }\zeta>1]\,(b).\end{aligned}\tag{34}$$

Condition (a) is that the mycorrhiza is less efficient by a factor ς (<1) for N uptake than the root [Eq. (32)], but the root is more efficient at P uptake than the mycorrhiza by a factor ξ which is greater than $1/\varsigma(>1)$. Overall the root has higher efficiency for both N and P uptake; the plant can grow faster without any mycorrhiza being present. Conversely, condition (b) is that the

mycorrhiza is more efficient by a factor $\varsigma(>1)$ for N uptake than the root, but the root is less efficient at P uptake than the mycorrhiza by a factor ξ which is less than $1/\varsigma(<1)$. Overall the mycorrhiza has higher efficiency for both N and P uptake; the plant can grow faster without any root being present.

In these cases the analysis terminating in the optimum allocation criteria of Eq. (30) breaks down, and it is necessary to redo the analysis for a plant with no mycorrhiza (or with no root). This is now done.

2.4. Optimum allocation in the absence of mycorrhiza

If, for instance, the uptake constants [Eq. (1)] are such that [cf. Eq. (32)–(34)]

$$k_{\mathrm{Nrt}} \geq k_{\mathrm{Nmy}} \quad \text{and} \quad k_{\mathrm{Prt}} \geq k_{\mathrm{Pmy}} \tag{35}$$

then clearly the plant can grow fastest without any mycorrhiza being present (Fig. 2B). To obtain optimum allocation in this case, we write the following. Eq. (18) with $\lambda_{\mathrm{my}}=0$ become

$$\begin{aligned} &\mu = k_{\mathrm{G}} C_{\mathrm{S}} N_{\mathrm{S}} P_{\mathrm{S}}, \\ &\mu(C_{\mathrm{S}} + f_{\mathrm{CX}}) = k_{\mathrm{C}} \lambda_{\mathrm{sh}}, \\ &\mu(N_{\mathrm{S}} + f_{\mathrm{NX}}) = k_{\mathrm{Nrt}} \lambda_{\mathrm{rt}}, \\ &\mu(P_{\mathrm{S}} + f_{\mathrm{PX}}) = k_{\mathrm{Prt}} \lambda_{\mathrm{rt}}, \\ &\lambda_{\mathrm{sh}} + \lambda_{\mathrm{rt}} = 1. \end{aligned} \tag{36}$$

Eliminating λ_{rt}:

$$\begin{aligned} &\mu(C_{\mathrm{S}} + f_{\mathrm{CX}}) = k_{\mathrm{C}} \lambda_{\mathrm{sh}}, \\ &\mu(N_{\mathrm{S}} + f_{\mathrm{NX}}) = k_{\mathrm{Nrt}}(1 - \lambda_{\mathrm{sh}}), \\ &\mu(P_{\mathrm{S}} + f_{\mathrm{PX}}) = k_{\mathrm{Prt}}(1 - \lambda_{\mathrm{sh}}), \\ &\mu = k_{\mathrm{G}} C_{\mathrm{S}} N_{\mathrm{S}} P_{\mathrm{S}}. \end{aligned} \tag{37}$$

Differentiating with respect to λ_{sh} and denoting differentials $\mathrm{d}x/\mathrm{d}\lambda_{\mathrm{sh}}$ by x':

$$\begin{aligned} &\mu' C_{\mathrm{S}} + \mu' f_{\mathrm{CX}} + \mu C_{\mathrm{S}}' = k_{\mathrm{C}}, \\ &\mu' N_{\mathrm{S}} + \mu' f_{\mathrm{NX}} + \mu N_{\mathrm{S}}' = -k_{\mathrm{Nrt}}, \\ &\mu' P_{\mathrm{S}} + \mu' f_{\mathrm{PX}} + \mu P_{\mathrm{S}}' = -k_{\mathrm{Prt}}, \\ &\mu' = k_{\mathrm{G}}(C_{\mathrm{S}}' N_{\mathrm{S}} P_{\mathrm{S}} + C_{\mathrm{S}} N_{\mathrm{S}}' P_{\mathrm{S}} + C_{\mathrm{S}} N_{\mathrm{S}} P_{\mathrm{S}}'). \end{aligned} \tag{38}$$

Maximum growth rate μ is obtained when $\mu'=0$ (using $\tilde{x}$ to denote the value of variable x at the maximum):

$$\begin{aligned} &\frac{\tilde{\mu}\tilde{C}_{\mathrm{S}}'}{k_{\mathrm{C}}} = 1, \quad \frac{\tilde{\mu}\tilde{N}_{\mathrm{S}}'}{k_{\mathrm{Nrt}}} = -1, \quad \frac{\tilde{\mu}\tilde{P}_{\mathrm{S}}'}{k_{\mathrm{Prt}}} = -1, \\ &0 = \tilde{C}_{\mathrm{S}}' \tilde{N}_{\mathrm{S}} \tilde{P}_{\mathrm{S}} + \tilde{C}_{\mathrm{S}} \tilde{N}_{\mathrm{S}}' \tilde{P}_{\mathrm{S}} + \tilde{C}_{\mathrm{S}} \tilde{N}_{\mathrm{S}} \tilde{P}_{\mathrm{S}}'. \end{aligned} \tag{39}$$

Using the first three equations to substitute for $\tilde{C}_{\mathrm{S}}'$, $\tilde{N}_{\mathrm{S}}'$ and $\tilde{P}_{\mathrm{S}}'$ in the last equation [cf. Eq. (30)]:

$$\frac{k_{\mathrm{C}}}{\tilde{C}_{\mathrm{S}}} = \frac{k_{\mathrm{Nrt}}}{\tilde{N}_{\mathrm{S}}} + \frac{k_{\mathrm{Prt}}}{\tilde{P}_{\mathrm{S}}} \tag{40}$$

This result was obtained in a solution to an exercise in Thornley and Johnson (2000, p. 630, Eq. (S13.5e)).

Teleonomic allocation in the absence of mycorrhiza is achieved by [cf. Eq. (31)]

$$\begin{aligned} &\lambda_{\mathrm{sh}}(\text{teleonomic}) = \frac{f_{\mathrm{Xsh}}(k_{\mathrm{C}}/C_{\mathrm{S}})^q}{f_{\mathrm{Xsh}}(k_{\mathrm{C}}/C_{\mathrm{S}})^q + f_{\mathrm{Xrt}}[(k_{\mathrm{Nrt}}/N_{\mathrm{S}})^q + (k_{\mathrm{Prt}}/P_{\mathrm{S}})^q]}, \\ &\lambda_{\mathrm{rt}}(\text{teleonomic}) = \frac{f_{\mathrm{Xrt}}[(k_{\mathrm{Nrt}}/N_{\mathrm{S}})^q + (k_{\mathrm{Prt}}/P_{\mathrm{S}})^q]}{f_{\mathrm{Xsh}}(k_{\mathrm{C}}/C_{\mathrm{S}})^q + f_{\mathrm{Xrt}}[(k_{\mathrm{Nrt}}/N_{\mathrm{S}})^q + (k_{\mathrm{Prt}}/P_{\mathrm{S}})^q]}, \\ &\lambda_{\mathrm{my}}(\text{teleonomic}) = 0, \\ &q = 1. \end{aligned} \tag{41}$$

When Eq. (40) is satisfied, then Eq. (41) give a result that satisfies Eq. (17).

Because the model (Fig. 1) is equivalent with respect to root (rt) and to mycorrhiza (my), this analysis applying to the system without a mycorrhiza can be applied to the system without the root by replacing the subscript "rt" with the subscript "my" and vice versa in Eqs. (36)–(41). This occurs if [cf. Eq. (35)]

$$k_{\mathrm{Nrt}} \leq k_{\mathrm{Nmy}} \quad \text{and} \quad k_{\mathrm{Prt}} \leq k_{\mathrm{Pmy}}. \tag{42}$$

If both Eqs. (35) and (42) are satisfied (by the equality) then root and mycorrhiza are equally effective at both N and P uptake and root: mycorrhiza allocation does not affect growth rate (Fig. 2C).

2.5. Numerical methods

Programming was in Advanced Continuous Simulation Language (ACSL, Aegis Research, Huntsville, AL, USA; version 11.8.4), a fortran-based ordinary differential equation (ODE) solver. The ODE's are integrated using Euler's method with a fixed integration interval of $\Delta t=0.001$ or 0.0001 d (where this was required for stable integration). A programme listing can be obtained from the first author.

3. Results

The model (Fig. 1) can each be exercised in two modes: with fixed allocation parameters (as in Fig. 3) or with allocation parameters which self-adjust to give maximum growth rate [Fig. 4; Eqs. (11), (31), and (41)]. Fixed allocation is not realistic for different environments, here represented by the uptake parameters [Eq. (1)]. Teleonomically-adjusted allocation fractions which maximize growth rate are our principal concern, and these are, relatively, "objective". Many of the responses are similar, and therefore we give some that are typical or of special interest.

3.1. Constant dynamic allocation parameters [Eq. (11)]

Fig. 3 illustrates the dynamics without teleonomy. There is a transient period lasting some 30 d after which exponential growth takes hold with all intensive quantities being constant (Fig. 3A and B). Proportional growth rate in exponential growth is 0.200 d^{-1} [Fig. 3B, Eq. (14)]. In equilibrium growth, allocation and partitioning fractions are equal [Eqs. (11) and (17)] at 0.4 (shoot), 0.35 (root) and 0.25 (mycorrhiza) (Fig. 3C); the partitioning fractions [Eq. (4)] approach the constant allocation fractions [Eq. (11)] asymptotically without overshoot. Because the shoot allocation fraction, at $\lambda_{\mathrm{sh}}=0.4$, is far below the optimum value of 0.74 (below and Fig. 4C), C substrate concentration, C_{S}, is very low (Fig. 3C). Fig. 3C shows how the allocation fractions lead the structural fractions to an extent which depends on the value of q (not shown). The most unstable components of the system are the substrate concentrations (C_{S}, N_{S} and P_{S}, Fig. 3D) which all overshoot before settling down to values of 0.00027, 0.099 and 0.038 kg substrate C, N, P (kg structural dry matter)$^{-1}$ [Fig. 3D, Eq. (2)]. The results here depend on the particular values assigned to the constant allocation parameters [Eq. (11)], and of course on the structure of the model, where possibly the lack of capacitance and resistance components allows these instabilities.

3.2. Teleonomic dynamic allocation parameters [Eqs. (11) and (31)]

Next the response with teleonomically self-adjusting allocation coefficients is examined [Eqs. (11) and (31)]. Fig. 4 shows two noteworthy features when compared with Fig. 3 which has fixed allocation parameters. First, the teleonomic algorithm [Eq. (31)] gives rise to pronounced damped oscillations (Fig. 4B, C, and D) (Note that the system has deliberately been started far from equilibrium as this reveals most about system dynamics – this

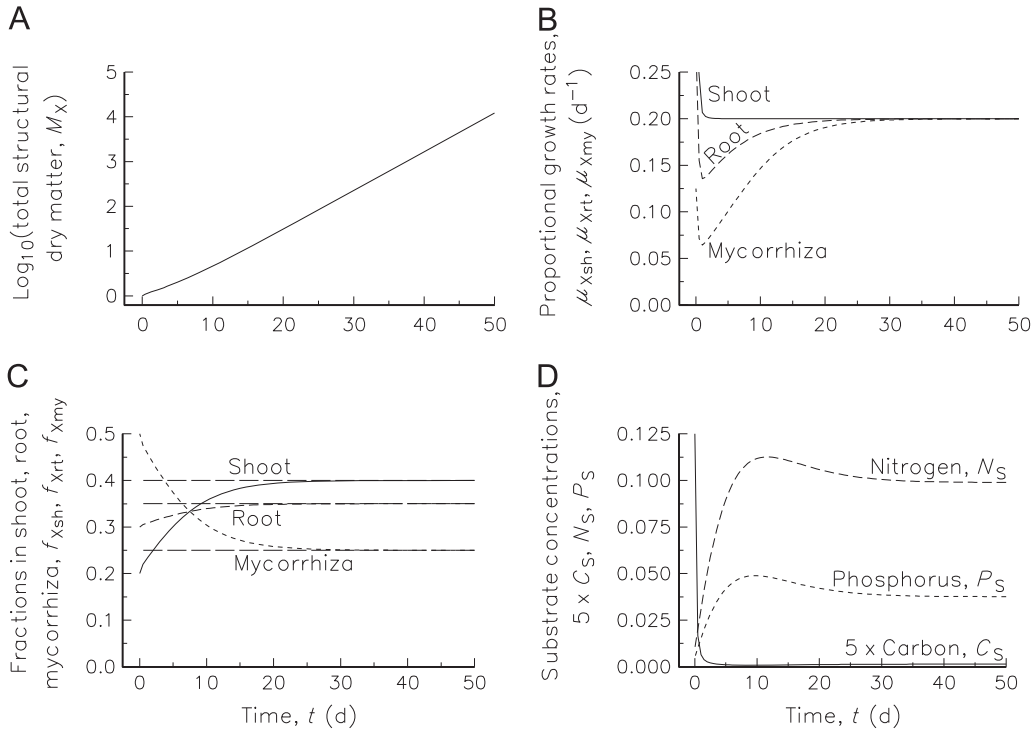


Fig. 3. Dynamics of default model without teleonomy. The allocation parameters, λ_{sh}, λ_{rt}, and λ_{my}, (shoot, root, mycorrhiza), are constant at 0.4, 0.35, and 0.25, (Eq. (11), Table 1, and Fig. 1). Plotted variables are: (A) total structural dry matter, M_X, Eq. (3); (B) structural proportional growth rates, Eq. (14); (C) shoot, root and mycorrhiza structural fractions, Eq. (4) – the dashed asymptotes show the constant values of the allocation parameters; (D) substrate concentrations of C, N and P, Eq. (2).

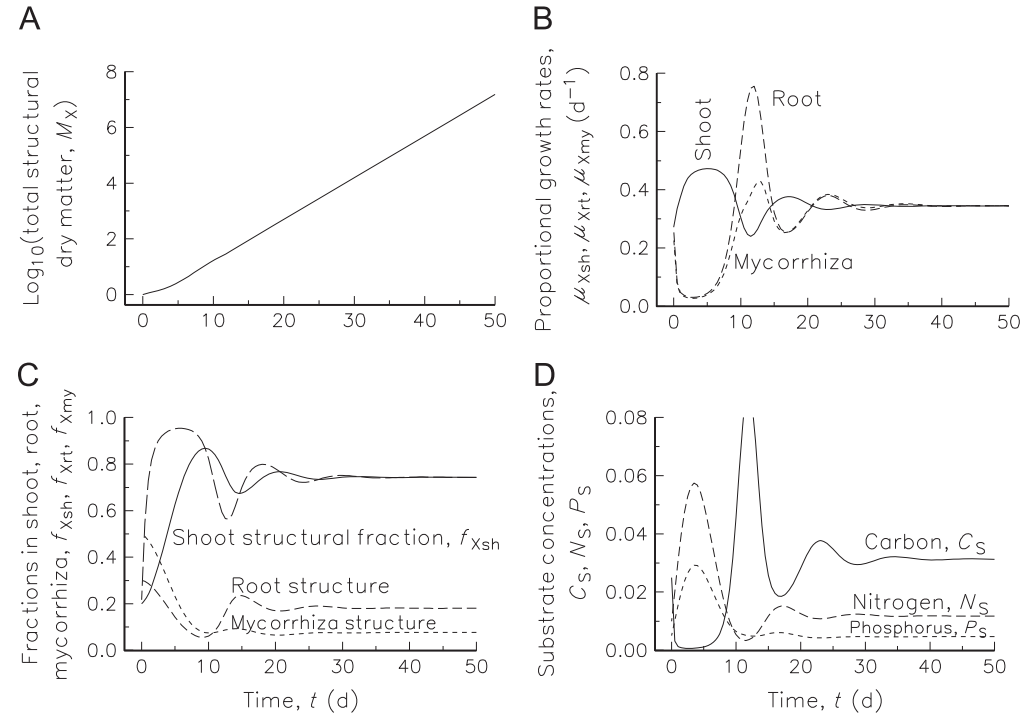


Fig. 4. Dynamics of default model with teleonomy – allocation parameters, λ_{sh} (shoot), λ_{rt} (root) and λ_{my} (mycorrhiza) self-adjust to give maximum proportional growth rate [Eqs. (11) and (31)]. Other parameters have their default values (Table 1). Plotted variables are: (A) total structural dry matter, M_X, Eq. (3); (B) structural proportional growth rates, Eq. (14); (C) shoot, root and mycorrhiza structural fractions, Eq. (4) – plus the teleonomic shoot allocation fraction, λ_{Xsh} [Eq. (31)]; (D) substrate concentrations of C, N and P, Eq. (2).

has the effect of emphasizing instabilities). These are most evident for substrate concentrations (Fig. 4D), with C (the principal substrate) > N > P. Second, proportional growth rate [Eq. (14)] has increased from 0.200 to 0.345 d^{-1}, a substantial increase, reflecting the fact that the fixed default allocation parameters [Eq. (11)] used in Fig. 3 (λ_{sh}=0.40, λ_{rt}=0.35 and λ_{my}=0.25) are a considerable distance from the optimized parameters (λ_{sh}=0.74, λ_{rt}=0.18 and λ_{my}=0.08) finally attained in Fig. 4, when Eq. (28) are satisfied. Fig. 3C compares the shoot allocation fraction λ_{sh} with the structural fraction f_{Xsh}. The allocation fraction leads the structural fraction with greater oscillatory amplitude. Root and mycorrhizal allocation fractions λ_{Xrt} and λ_{Xmy} (not shown) behave

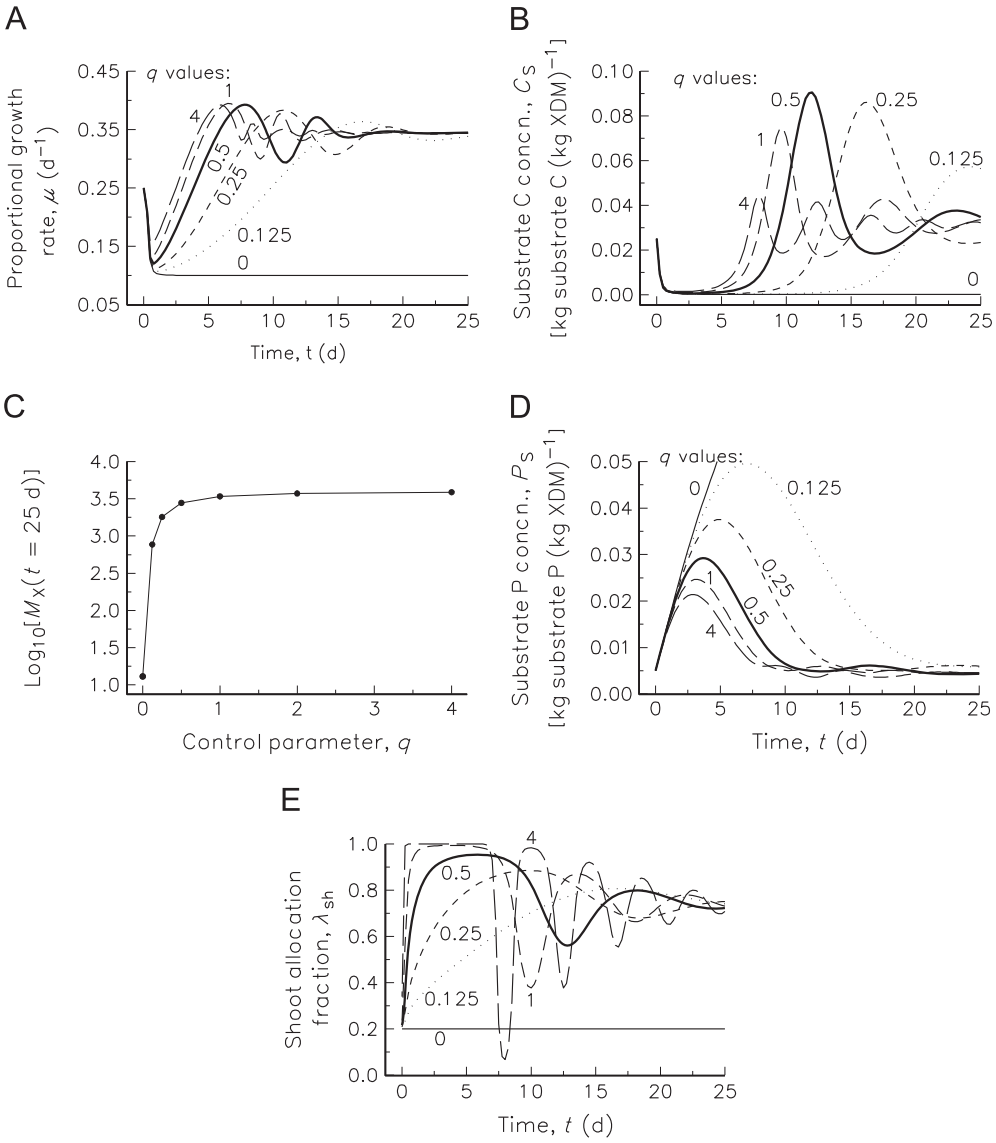


Fig. 5. Default model dynamics using teleonomic allocation [Eq. (11)], now for a range of values of the control parameter, q [Eq. (31)] as indicated in Fig. 5A, B, D. $q=0.5$ is the default value of q, shown by the continuous bold line. Other parameters have their default values (Table 1). Plotted variables are: (A) proportional growth rate, μ [Eq. (7)]; (B) substrate C concentration, C_S [Eq. (2)]; (C) $\log_{10}$ [structural dry matter, Eq. (3), at time $t=25$ d]; (D) substrate P concentrations, P_S, Eq. (2) (XDM denotes structural dry matter); and (E) shoot allocation fraction, λ_{sh} [Eq. (31)].

similarly with respect to the structural fractions f_{Xrt} and f_{Xmy}. "Steady-state" substrate concentrations, C_S, N_S and P_S, now take more realistic values of 0.031, 0.012 and 0.0047 kg substrate C, N, P (kg structural dry matter)$^{-1}$ [Fig. 4D, Eq. (2)].

To investigate the oscillations in Fig. 4, the teleonomic control parameter q [Eq. (31)] is varied over the range 0–4, the results being shown in Fig. 5. Eq. (31) with $q=0$ compels allocation fractions [λ, Eqs. (10) and (11)] to remain equal to current partition fractions [f_X, Eq. (4)], so the λ do not change during growth. This gives growth which is very sub-optimal (Fig. 5A, C, as shown also in Fig. 3). Increasingly positive values of q give convergence on the same equilibrium maximum proportional growth-rate solutions (Fig. 5A); convergence is faster for higher q's; this gives a dry matter value at a particular time (say at $t=25$ d, Fig. 5C, perhaps related to the length of the growing season) which increases asymptotically with q. Lower q values lower the frequency of oscillations and mostly increase their amplitude; however, lower q values than 0.5 can increase greatly the time taken to attain optimum allocation. Oscillations in allocation fractions are limited to the range (0, 1) whereas substrate concentrations have no upper limit (Fig. 5E). Note that for q values above about 0.5, the graphs tend to converge – this is the reason why, empirically, a $q=0.5$ is taken as the default value, although values of 1 or 2 may mechanistically be more realistic – integer values of q (e.g. 1, 2,…,) with a maximum value of 4 are favoured from mechanistic consideration of allosteric proteins (Tsai et al., 2009; Monod et al., 1965).

3.3. Teleonomic allocation with product inhibition of uptake

As noted by Johnson (1985), the instabilities seen in Figs. 4 and 5 are markedly decreased by introducing substrate inhibition of uptake flux into the uptake equations [Eq. (1)]. Product inhibition of uptake may also be biologically appropriate (see Eq. (7) of Johnson, 1985). Assume Eq. (1) are replaced by

$$U_C = \frac{k_C M_{Xsh}}{1+C_S/J_{CS}}, \quad U_N = \frac{k_N M_{Xrt}}{1+N_S/J_{NS}}, \quad U_P = \frac{k_P M_{Xmy}}{1+P_S/J_{PS}},$$
$$J_{CS} = 0.03 \text{ kg C substrate (kg shoot structural dry matter)}^{-1}, \quad (43)$$
$$J_{NS} = 0.01 \text{ kg N substrate (kg root structural dry matter)}^{-1},$$
$$J_{PS} = 0.005 \text{ kg P substrate (kg mycorrhizal structural dry matter)}^{-1}.$$

J_{CS}, J_{NS} and J_{PS} are inhibition parameters (Thornley and France, 2007, p. 115), with values that are the rounded values of the equilibrium substrate concentrations attained in Fig. 4D (optimization with default parameters). Assigning large values to the J parameters switches off inhibition (this is the default position, Table 1). Product inhibition is applied only in Fig. 6 to illustrate its potent impact on system dynamics. Note that optimization Eq. (28) are no longer valid when Eq. (1) are replaced by Eq. (43) – in this case we were unable to obtain analytical solutions to the optimization problem, but we confirmed that the solution shown in Fig. 6 does not maximize growth rate with respect to the allocation parameters [Eq. (10)]. An optimum could be found for the situation where Eq. (1) are replaced by Eq. (43) but this is no longer a matter of straightforward algebra and would involve numerical solutions. However, we used Eqs. (28) and (31) to generate the results shown in Fig. 6.

The oscillations of Fig. 5 are much decreased. Before condemning the teleonomic method on account of these instabilities, it should be noted that a comparison of teleonomic and resistance allocation models without uptake inhibition suggests that similar instabilities occur in the mechanistic resistance approach (Thornley, 1995, Fig. 3), and may therefore be an irreducible feature of allocation and growth processes, which combine transport and chemical conversion. However, as mentioned above, the inclusion of more resistance and capacitance components in both models (teleonomic as here and mechanistic) may decrease oscillations as well as being more realistic.

Growth is also considerably decreased when Eq. (43) replace Eq. (1), so that total structural dry matter, M_X, [Eq. (3) at time $t=25$ d] is now 123 kg compared to 3417 kg without inhibition. This is to be expected as Eq. (43) will roughly halve the substrate uptake fluxes.

3.4. Responses to substrate supplies

Next "environmental" responses are examined: the supplies of C, N and P substrates are altered by changing the uptake parameters, k_C, k_{Nrt}, k_{Pmy}, k_{Nmy}, k_{Prt} [Eq. (1), Fig. 1]. These results are obtained using the default model without product inhibition of uptake [Eq. (43)], and applying teleonomically optimized allocation coefficients, λ_{sh}, λ_{rt} and λ_{my} [Fig. 1, Eqs. (10), (11), (31), and (41)].

The simplest of the five responses to the uptake rate constants [Eq. (1)] is that to increasing C uptake, k_C, drawn in Fig. 7A, D, G. For much of the response (Fig. 7A), the growth rate is C-limited [e.g. Eqs. (20) and (21)]; if plotted linearly (μ:k_C), there are diminishing returns to increases in shoot C uptake. Shoot fraction decreases whereas both root and mycorrhizal fractions increase (Fig. 7D). C substrate concentration increases while again both N and P substrate concentrations decrease. This is a consequence of Eq. (6) for growth rate.

The responses to N substrate uptake by the root fraction, k_{Nrt}, are not so straightforward. Growth rate increases with increasing k_{Nrt} (Fig. 7B) but not as strongly as in Fig. 7A (our default k values correspond to C being the main limitation). At the two lowest values of k_{Nrt} (0.005, 0.015), Eq. (42) is satisfied ($0.015 < 0.025$, $0.01 < 0.02$) so that the plant can grow fastest without any root (Fig. 2D), giving the cross-over seen in Fig. 7E; the root fraction passes through a maximum. N substrate increases throughout (Fig. 7H) and P substrate decreases (as expected) whereas C substrate passes through a maximum. Such responses, if observed, might be difficult to interpret without a model.

Responses to P substrate uptake by the mycorrhiza (k_{Pmy}) are shown in Fig. 7C, F, and I. Growth rate increases slightly (Fig. 7C). At the two lowest values of k_{Pmy} (0.002, 0.006), Eq. (35) is satisfied ($0.05 > 0.01$, $0.01 > 0.006$) so that fastest growth occurs when the mycorrhiza is absent (Fig. 2B) giving the results in Fig. 7F; this time the mycorrhiza fraction passes through a maximum. The substrate concentrations (Fig. 7I) behave according to expectation [P increases, C and N decrease; Eq. (6)].

Fig. 8 gives similar responses to those shown in Fig. 7, but this time for varying the N substrate uptake rate of the mycorrhiza (k_{Nmy}) and the P substrate uptake rate of the root (k_{Prt}) [Eq. (1)]. Growth rate always increases with increasing uptake rate constant (Fig. 8A and B). At the highest values of mycorrhizal N uptake

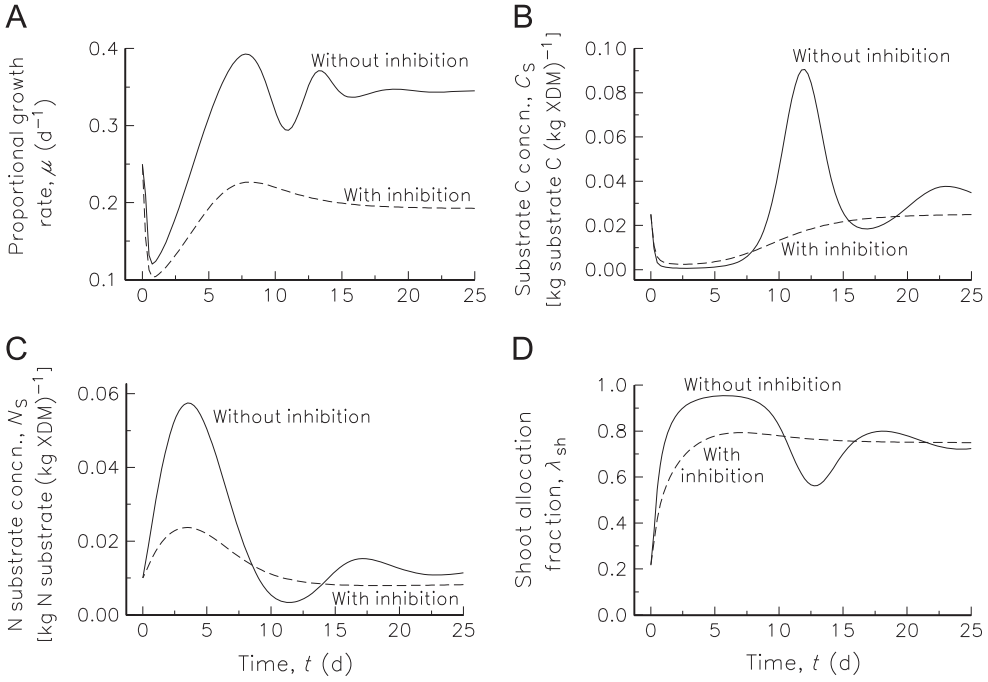


Fig. 6. Dynamics of the default model, but with uptake inhibition as in Eq. (43) which replace Eq. (1). Eqs. (11) and (31) are used to adjust the allocation parameters λ, although Eq. (31) are no longer valid when Eq. (43) replaced Eq. (1). Other parameters are default (Table 1). Plotted variables are: (A) proportional growth rate, μ [Eqs. (14) and (15)]; (B, C) Substrate concentrations of C and N, C_S and N_S, Eq. (2) (XDM denotes structural dry matter); and (D) shoot allocation fraction, λ_{sh} [Eq. (31)].

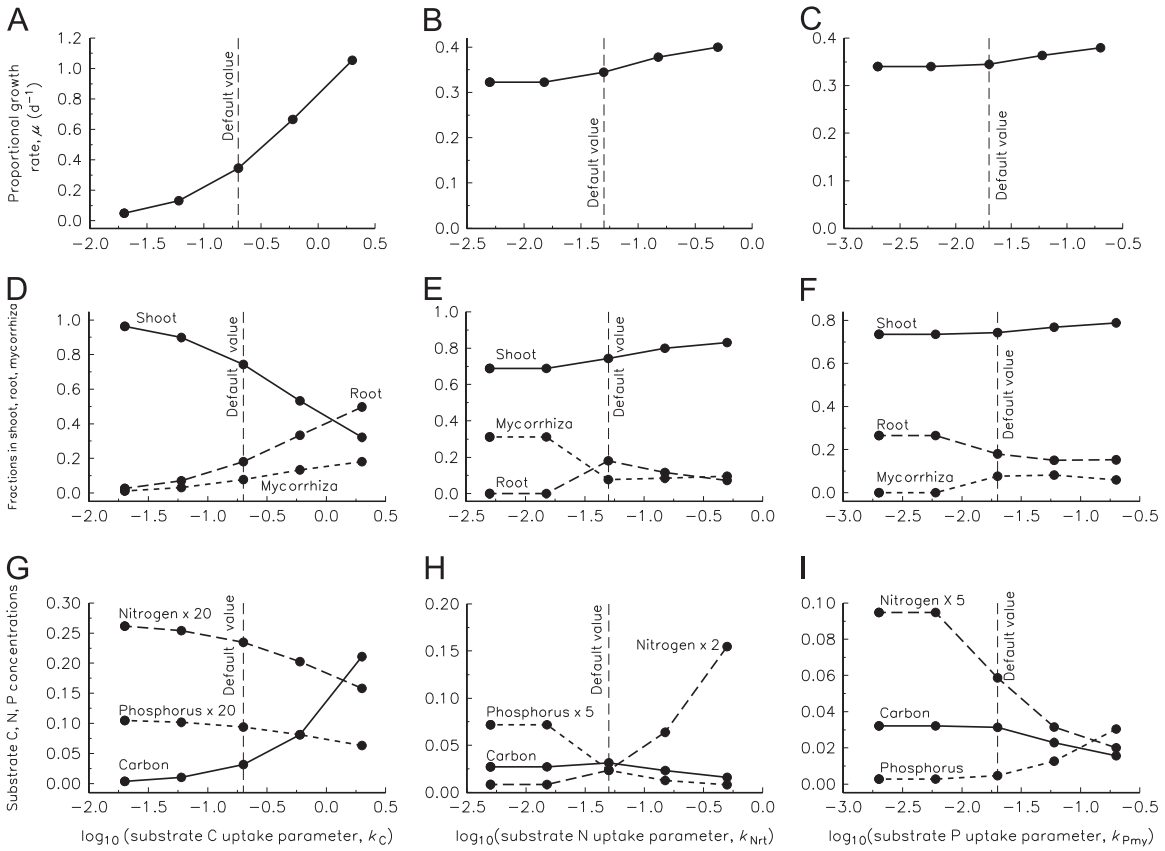


Fig. 7. Uptake rates, k_C, k_{Nrt} and k_{Pmy} are separately varied [Eq. (1)], each over a 100-fold range with all other parameters at default (Table 1). Teleonomy is applied – allocation parameters, λ_{sh} (shoot), λ_{rt} (root) and λ_{my} (mycorrhiza) self-adjust to give maximum proportional growth rate [Eqs. (11), (31), (41) and below]. Parts A, B, C show the effect on proportional rate of growth, μ [Eqs. (14) and (15)]; D, E, F on structural fractions in shoot, root and mycorrhiza [Eq. (4)]; G, H, I on substrate concentrations [Eq. (2)], with units of kg substrate C, N, P (kg structural dry matter)$^{-1}$. The default result is shown by the vertical dashed line.

(k_{Nmy}) there is no root (Figs. 8C and 2D), and at the highest values of root P uptake there is no mycorrhiza (Fig. 8D and 2B). The substrates behave in a more complex and counter-intuitive manner than in Fig. 7H and I. Example in Fig. 8E, an initial increase in the mycorrhizal N uptake parameter causes N substrate to decrease and P substrate to increase. This appears to result from the altered root-mycorrhiza fractions, the differing uptake rates of root and mycorrhiza for N and P uptakes and the growth equation [Eq. (6)]. Similar behaviour is seen in Fig. 8F where responses to root P substrate uptake (k_{Prt}) are drawn.

3.5. Responses to utilization rate, k_G

First it may be noted that the utilization constant, k_G, in Eq. (6) does not intrinsically favour either of the three substrates, and nor is it directly instrumental in allocation between shoot, root and mycorrhiza. Fig. 9A shows how the proportional growth rate (pgr), μ [Eqs. (7) and (14)], in equilibrium exponential growth is always below the three asymptotes given in Eq. (20) and also drawn in Fig. 9A. At high k_G where the substrates become a negligible component of growth (Fig. 9C) (that is, growth is almost entirely growth in structural dry matter and growth in the substrate pools is of minor significance), the pgr approaches the lowest (limiting) asymptote very closely. In Fig. 9B it can be seen that increasing k_G (which could be due to increased temperature) shifts the allocation fractions [Eq. (10)] favouring the system component which is the more limiting [in this case the shoot, Eq. (20)]. Last, Fig. 9C illustrates how increasing k_G decreases all the substrate concentrations, but decreases the limiting substrate (substrate C) least because of the increasing shoot fraction (Fig. 9B).

4. Discussion

The model of Fig. 1 and subsequent analysis are proposed as a possible method of simulating the highly variable allocation fractions which are encountered in many crops, plants and plant ecosystems. In particular we consider the question of resource allocation "outside" a single species (here a plant) and to another organism in an evolved multi-species association. For this we use a teleonomic approach (Monod, 1974; Reynolds and Thornley, 1982; Johnson and Thornley, 1987; Thornley and France, 2007). The principal strengths of teleonomy are that it provides a way of modelling allocation with fewer parameters compared to more mechanistic approaches, which, happily, it appears to simulate rather well (Mäkelä and Sievänen, 1987). Moreover, the approach offers a formal means to consider situations, such as a plant/fungal association, where there may be separate and even different mechanisms for control of growth within each species, but where a common overall goal, such as maximizing the benefit of the association, as here, might be assumed.

Five principal weaknesses, which apply to all optimal function models, are (1) some goal must be chosen and this is subjective (Cheeseman, 1993, remarked that "growth and resource allocation can be based solely on local, internal, …, rules, without the need to integrate activities at the organismal level"); (2) it seems likely that a goal such as proportional growth rate (as used here) may not apply to many situations where factors such as drought resistance or the ability to withstand predation may be more important; (3) the response curve being optimized may be fairly flat (Fig. 2) – which is to say that the particular allocation fractions may not affect plant growth that much, and may therefore be

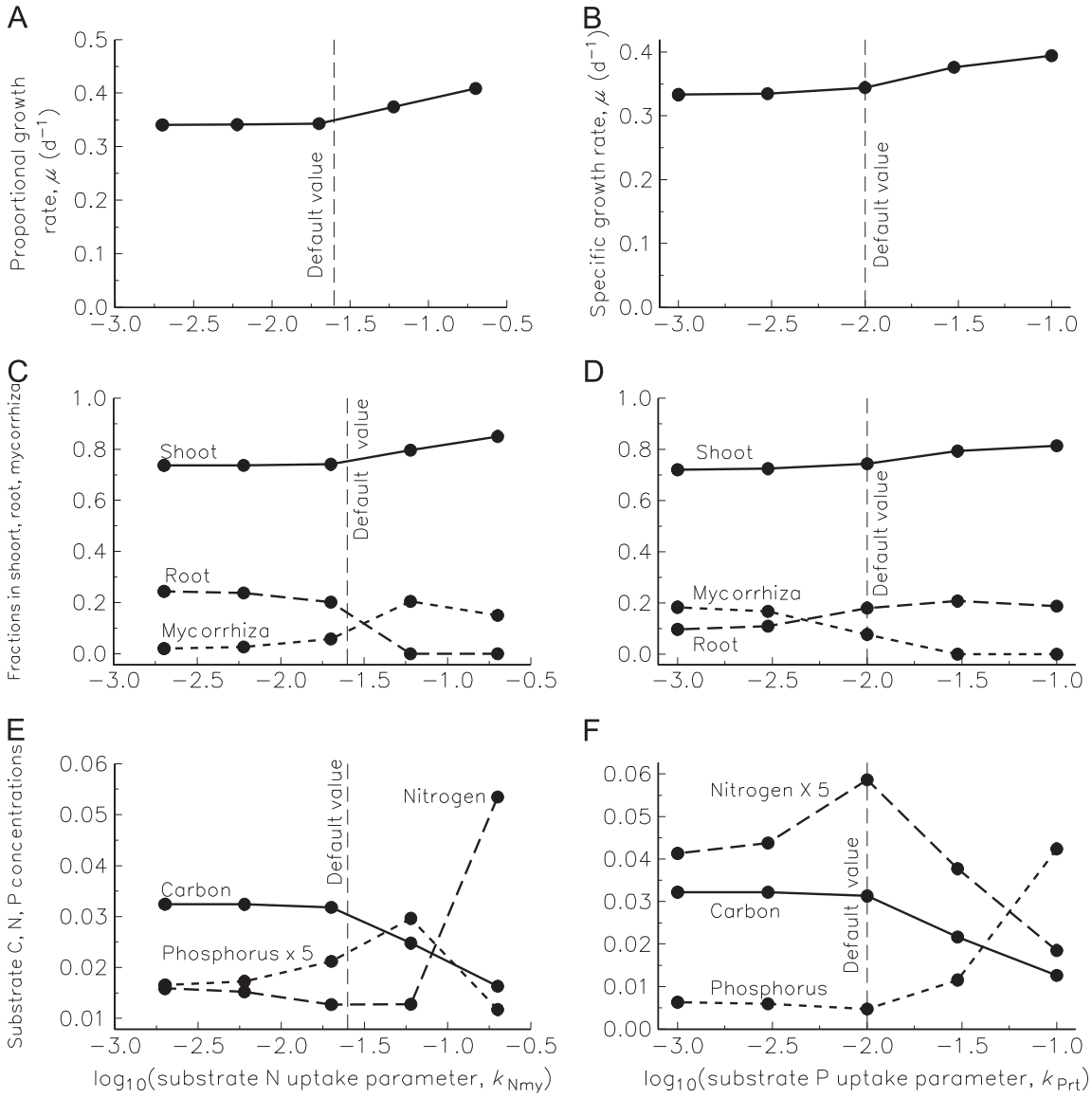


Fig. 8. Uptake rates, k_{Nmy} and k_{Prt} are separately varied [Eq. (1)], each over a 100-fold range with all other parameters at default (Table 1). Teleonomy is applied – allocation parameters, λ_{sh} (shoot), λ_{rt} (root) and λ_{my} (mycorrhiza) self-adjust to give maximum proportional growth rate [Eqs. (11), (31), (41) and below]. Parts A, B show the effect on proportional rate of growth, μ [Eqs. (14) and (15)]; C, D on structural fractions in shoot, root and mycorrhiza [Eq. (4)]; E, F on substrate concentrations [Eq. (2)], with units of kg substrate C, N, P (kg structural dry matter)$^{-1}$. The default result is shown by the vertical dashed line.

variable but this variability is not significant; (4) the last item possibly causes the system to suffer from damped oscillations (Fig. 4) – which may not be realistic, although the variability and contrariness of measurements of allocation (Wilson, 1988) would not rule this out, and it may be noted that a comparison of teleonomic and resistance allocation models without uptake inhibition suggests that similar instabilities occur in the mechanistic resistance approach (Thornley, 1995, Fig. 3); and finally (5) a teleonomic model is always an incomplete statement – a mechanistic statement of how the teleonomy arises is more complete, and although more complex, it offers more possibilities and may be easier to work with.

It may be that teleonomic allocation has not fulfilled its early promise, in that two early users of the method have decided on alternative approaches. In his grassland ecosystem model (Thornley, 1998b), teleonomy was jettisoned in favour of the more transparent mechanistic transport-resistance approach [remember that, as shown by Mäkelä and Sievänen (1987), this gives an apparently teleonomic response]. Indeed, the feasibility of mechanistic modelling is improved by simpler representations of mechanisms such as suggested by Pickard (2012). Johnson et al. 2010, Eq. (20) choose a simple empirical dependence on ambient CO_2 concentration, ignoring other influences. Others have preferred to take substrates entirely out of the model by assuming that uptake of substrate (often on a daily basis but sometimes over a different time interval such as a phyllochron) is allocated and used up (e.g. Gabrielle et al., 2002; Christophe et al., 2008). Empirical and difficult-to-define concepts such as "sink strength" or "demand" or a "developmental index" may be used for this purpose. Sometimes these devices introduce circularity into the scheme. Arguably, this makes it difficult to represent the large changes in substrate concentrations which occur on a daily and seasonal basis, which greatly influence plant growth and development.

The above allocation model attends specifically to the question of resource sharing between shoot, root and associated mycorrhizae. Predictions are made to explore the insights that can be gained from this simple "stand-alone" system. Clearly other insights can be anticipated when such an algorithm is integrated into an ecosystem model where a more extensive range of limitations and requirements is considered. Solutions predicting

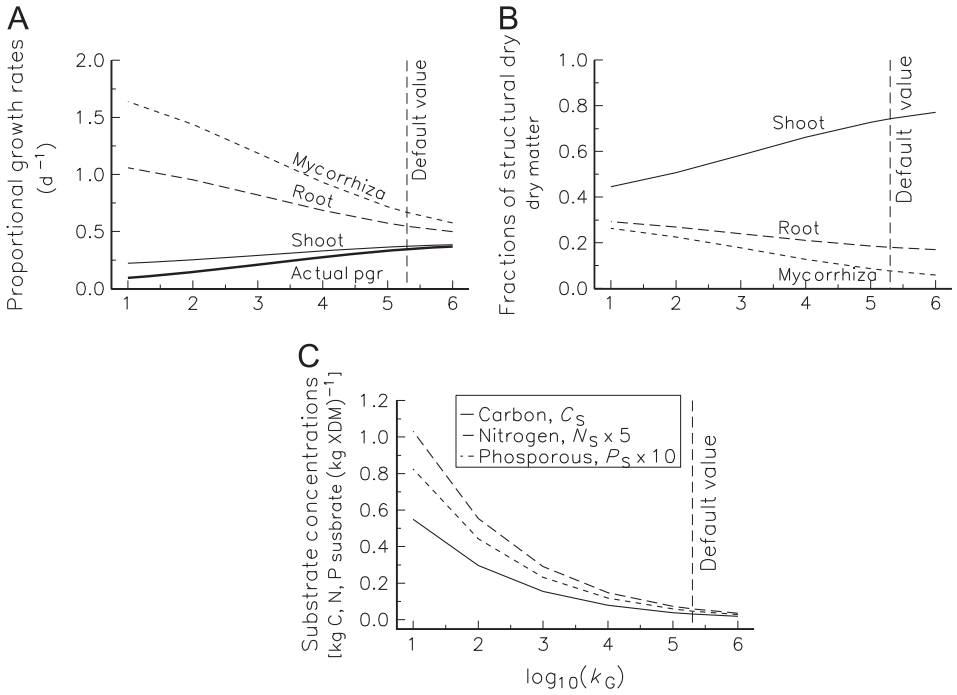


Fig. 9. Responses to utilization rate parameter k_G [Eq. (6)] are depicted, employing teleonomy to maximize growth rate and with all other parameters at default (Table 1). XDM denotes structural dry matter; pgr denotes proportional growth rate. (A) compares the actual pgr, μ [Eqs. (7) and (14)], in equilibrium growth with the limiting pgr values given by Eq. (20); (B) the fractions of system XDM in the parts of the system [Eq. (4) and Fig. 1] are drawn; (C) illustrates how the system substrate concentrations [Eq. (2)] depend on k_G. The dashed vertical lines show the position of the default value of k_G [Eq. (6)].

that growth is maximum in the absence of root are likely to be seen as strongly sub-optimal when anchorage or water uptake are also taken into account. Solutions predicting that growth is maximum in the absence of allocation to mycorrhizae, however, may also be of value. Pasture plants sown into recently ploughed soils in which mycorrhizal networks will have been disrupted, show a period of faster growth than is obtained in undisturbed soils (e.g. Parsons et al., 2011). Nevertheless, allocation of resources to mycorrhizae, even if sub-optimal for plant growth in the short-term, may be optimal in the longer term if there is uncertainty in the supply of a critical nutrient. A model of the kind proposed here provides scope for predicting changing priorities for allocation at the ecosystem scale.

There is substantial interest at present in the role of interactions between major food-producing plant species with soil organisms, particularly soil fungi. This is particularly so in highly productive cool-temperate grasslands, partly due to their major role in food production, and partly due to concern to maintain the large quantities of carbon sequestered to-date in such soils, which are facing increasing intensity of utilization and nutrient inputs (OECD/FAO, 2012). This has led to a reconsideration of the role of mycorrhizae in soil function (Treseder and Holden, 2013; Clemmensen et al., 2013; Treseder et al., 2012; Talbot and Treseder, 2011). Some valuable progress has been made (Orwin et al., 2011), but the question of just how much sharing of resources (e.g. through exudation of sugars and proteins into the rhizosphere, through 'in tissue' exchange in arbuscular mycorrhizae, and even through litter fall) can occur remains a substantial challenge. The above model offers one approach to these issues, although much remains to be done.

5. Conclusions

First, teleonomic or optimal response has a role in simplified models which may be constructed because (a) mechanistic models are too difficult with current knowledge, and (b) one wishes to explore the consequences of assuming certain fitness functions. However, they are subjective, to an extent arbitrary and cannot be validated (the ability to describe data cannot be viewed as a proper validation). Nevertheless, they can facilitate progress in tackling important problems, and may give rise to useful insights. The model presented above is to be viewed in this light.

Second, it should always be remembered that a mechanistic model is the ideal statement. It is objective, free-running, contains no explicit statement of goals although its predictions may appear to satisfy certain goals. Mäkelä and Sievänen (1987) showed valuably that teleonomic and mechanistic models of allocation can give virtually identical predictions. A mechanistic model is objective, in that the assumptions can and should be based on empirical data at the level of the assumed mechanisms. It can and should be tested in terms of its predictions and when there is a failure, this should be pinned down at the assumptions level. The fact that the mechanistic approach is able to give predictions which are apparently goal-seeking might provide some solace for teleonomically-minded ecologists.

Third, remembering that an optimal function may not be well-defined (as in this case: Fig. 2A) and therefore not very effective in "controlling" growth processes, may cause one to question the value of searching for optimal functions which are the product of evolution over millennia with unknown environmental and other constraints. Such scepticism may be supported by considering the numerous cases where evolution has led to poorly engineered solutions.

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